

# Fertilizer Recommendation System

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# 1 Executive Summary

## Overview

This project will recommend the optimal fertilizers for crops under various soil and environmental conditions. Machine learning technology is used to generate a model that helps predict the right fertilizer based on soil properties, nutrient levels, weather conditions, or crop types of their land information, ensuring the right fertilizer is used so that the farmers can get benefit from its use.

## Problem statement

Many farmers go with personal experience rather than relying on scientific evidence while choosing fertilizer. Such a choice leads to excessive usage of fertilizers since people have no idea about different environmental factors that can affect them, causing pollution in the process. Conversely, if farmers use fertilizers inadequately, it will lead to a reduced productivity level of crops.

## Proposed Solution

We suggest that a web-based machine learning system that can suggest fertilizers as per real-time and user-entered attributes such as temperature, moisture, soil type, crop type, soil pH, rain falls, season and NPK values. A classification is performed with a Support Vector Machine (SVM) model. Geolocation is used to retrieve real-time temperature automatically using the OpenWeather API.

## Methodology

The proposed system will implement an intelligent Fertilizer Recommendation System with the aid of a Support Vector Machine (SVM). In its development process, the system will use the model to train the SVM classifier through some agriculture data. The system's implementation will use a Flask backend that will do the model processing and predictions, while a frontend will facilitate easy interaction with the model inputs.

## Expected Outcomes

The resulting model proves to be highly accurate and is operationalized as an easy-to-use web app using Streamlit, enabling farmers and agricultural experts to get instant fertilizers based on inputting basic information about their field.

# 2 Introduction

## Background

Fertilizers are critical in today's farming industry as a way of providing soil with nutrients that plants need for proper development. Nutrients that are normally provided using fertilizers include nitrogen (N), phosphorus (P), potassium (K), among others. The choice of fertilizer to use involves an array of factors such as soil properties, crop selection, climate, seasons, and irrigating processes.

In the field of chemical engineering, management of nutrients in agriculture is directly linked to mass transfer and reaction rates in soils and crops. Incorporation of machine learning in this

sector combines data science with agrochemical engineering.

### **Problem Statement**

Farmers do not possess the right equipment and knowledge to choose the right fertilizers for their specific crops and soil types. The existing solutions lack specificity and fail to take into account the specific environmental factors. As a result, manual decisions lead to inefficiencies, wastage, poor yields, and harm the environment.

### **Objectives:**

- To build an ML model to suggest the optimal fertilizers based on the crop type, soil nutrients (N,P,K), and the temperature and humidity of the weather conditions (temperature, humidity and season).
- To integrate the information of the weather using OpenWeatherMap API for location-wise recommendations.
- To evaluate the efficiency of different models such as SVM, logistic regression, XGBoost, etc.
- To deploy the model through a Flask API where a user can enter the soil and weather information and obtain real-time recommendations about the fertilizer.
- To encourage the use of AI in agriculture for sustainable farming.

### 3 Methodology

#### Data Source:

Kaggle

The dataset used for this project is fertilizer\_recommendation.csv, a custom-curated dataset containing 10,000 records and 20 columns representing soil properties, environmental conditions, crop information and historical data. The target variable is Recommended Fertilizer.

#### Original dataset columns:

Soil\_Type, Soil\_pH, Soil\_Moisture, Organic\_Carbon, Electrical\_Conductivity

Nitrogen\_Level, Phosphorus\_Level, Potassium\_Level

Temperature, Humidity, Rainfall

Crop\_Type, Crop\_Growth\_Stage, Season, Irrigation\_Type

Previous\_Crop, Region, Fertilizer\_Used\_Last\_Season, Yield\_Last\_Season

Recommended\_Fertilizer (Target)

#### Data Preprocessing:

##### Feature Selection

Seven columns have been excluded from the initial data because they are either irrelevant for the immediate purpose, could cause data leakage (Fertilizer\_Used\_Last\_Season, Yield\_Last\_Season), or are very specific to regions (Region, Previous\_Crop). The remaining twelve features have been listed below.

| Retained Feature | Type        | Description                                           |
|------------------|-------------|-------------------------------------------------------|
| Soil_pH          | Numerical   | Acidity/alkalinity of the soil (0-14 scale)           |
| Soil_Moisture    | Numerical   | Soil water content as percentage                      |
| Nitrogen_Level   | Numerical   | Soil nitrogen concentration (mg/kg)                   |
| Phosphorus_Level | Numerical   | Soil phosphorus concentration (mg/kg)                 |
| Potassium_Level  | Numerical   | Soil potassium concentration (mg/kg)                  |
| Temperature      | Numerical   | Ambient temperature in degrees Celsius                |
| Humidity         | Numerical   | Relative humidity in percentage                       |
| Rainfall         | Numerical   | Annual rainfall in millimetres                        |
| Soil_Type        | Categorical | Clay, Silt, Sandy, or Loamy                           |
| Crop_Type        | Categorical | Cotton, Maize, Wheat, Potato, Rice, Sugarcane, Tomato |
| Season           | Categorical | Kharif, Zaid, or Rabi                                 |
| Irrigation_Type  | Categorical | Canal, Sprinkler, Rainfed, or Drip                    |

## Data Quality Checks

- **Missing Values:** There were no missing values in any of the twelve remaining variables.
- **Infinite Values:** There were no infinite values present after changing all the values to nan and then performing the check.
- **Duplicates:** The duplicates were determined by checking `df.duplicated().sum()`.
- **Whitespace:** The column names were stripped of whitespaces.

## Encoding Categorical Features

One Hot Encoding with `drop_first=True` was performed on all four categorical variables to prevent multicollinearity or dummy variable trap. This led to an increase from 12 input variables to 22 numerical variables.

| Categorical Feature           | Categories                      | Encoded Columns Created                                 |
|-------------------------------|---------------------------------|---------------------------------------------------------|
| Soil_Type (base: Clay)        | Clay, Silt, Sandy, Loamy        | Soil_Type_Loamy, Soil_Type_Sandy, Soil_Type_Silt        |
| Crop_Type (base: Cotton)      | 7 crops                         | Crop_Type_Maize, Potato, Rice, Sugarcane, Tomato, Wheat |
| Season (base: Kharif)         | Kharif, Zaid, Rabi              | Season_Rabi, Season_Zaid                                |
| Irrigation_Type (base: Canal) | Canal, Sprinkler, Rainfed, Drip | Irrigation_Type_Drip, Rainfed, Sprinkler                |

## Feature Scaling

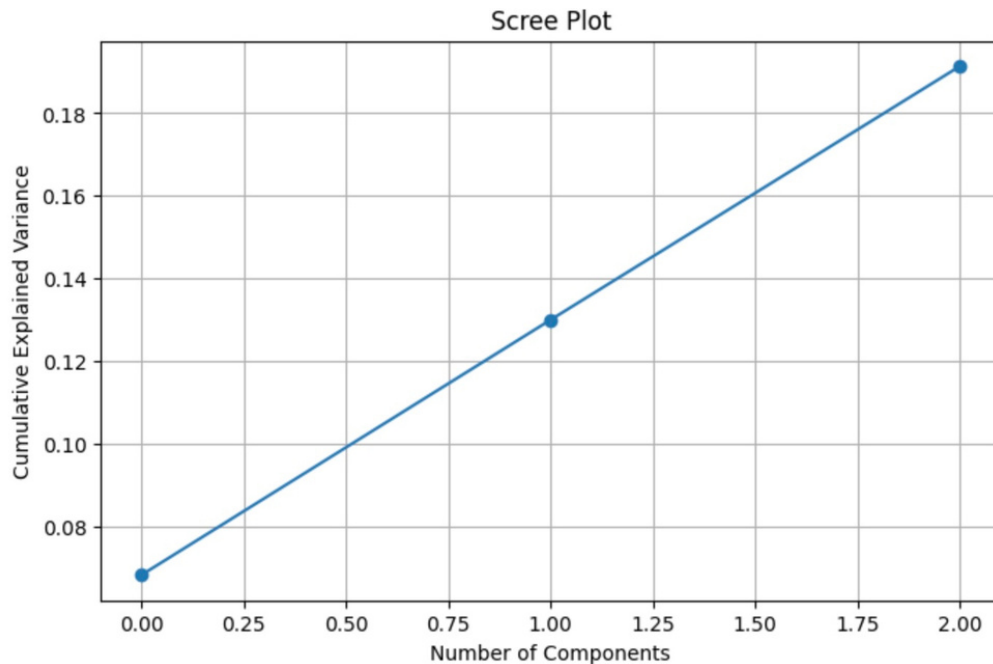
All the 22 features were scaled using `StandardScaler` in `scikit-learn`, with the purpose of having zero mean and standard deviation equal to one for each feature. This step is necessary because SVM, KNN, and logistic regression models require that all input features be scaled.

## Label Encoding

The `LabelEncoder` was used to encode the label of the target variable `Recommended_Fertilizer` to numerical values ranging from 0 to 6.

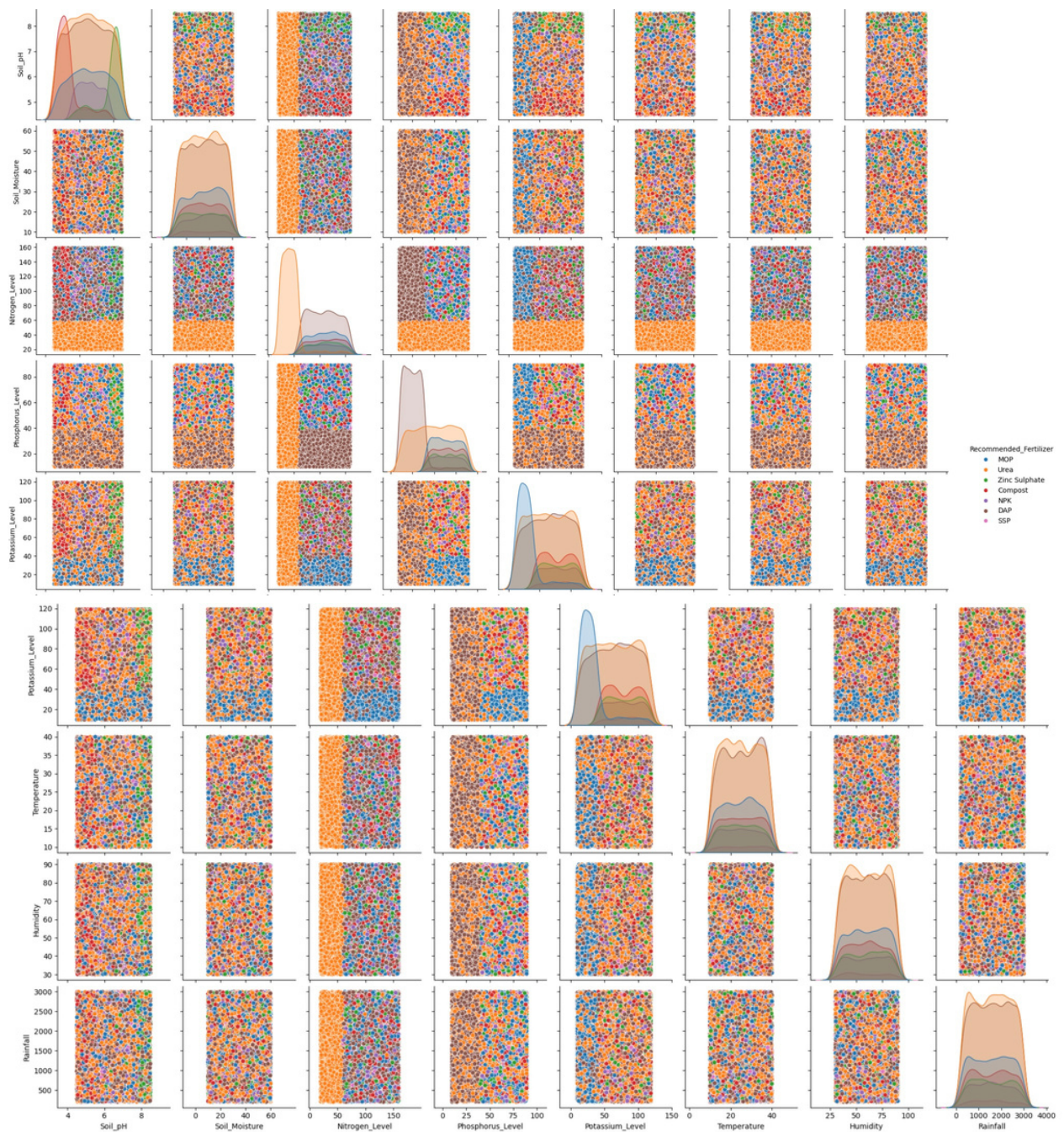
| Encoded Label | Fertilizer Name              | Primary Use                                    |
|---------------|------------------------------|------------------------------------------------|
| 0             | MOP (Muriate of Potash)      | Potassium source; used for root crops, fruits  |
| 1             | Urea                         | Nitrogen source; most widely used N fertilizer |
| 2             | Zinc Sulphate                | Micronutrient supplement for zinc deficiency   |
| 3             | Compost                      | Organic matter; improves soil structure        |
| 4             | NPK                          | Balanced macro-nutrient blend                  |
| 5             | DAP (Di-ammonium Phosphate)  | Nitrogen and phosphorus source; widely used    |
| 6             | SSP (Single Super Phosphate) | Phosphorus and sulphur source                  |

## Exploratory Data Analysis (EDA)

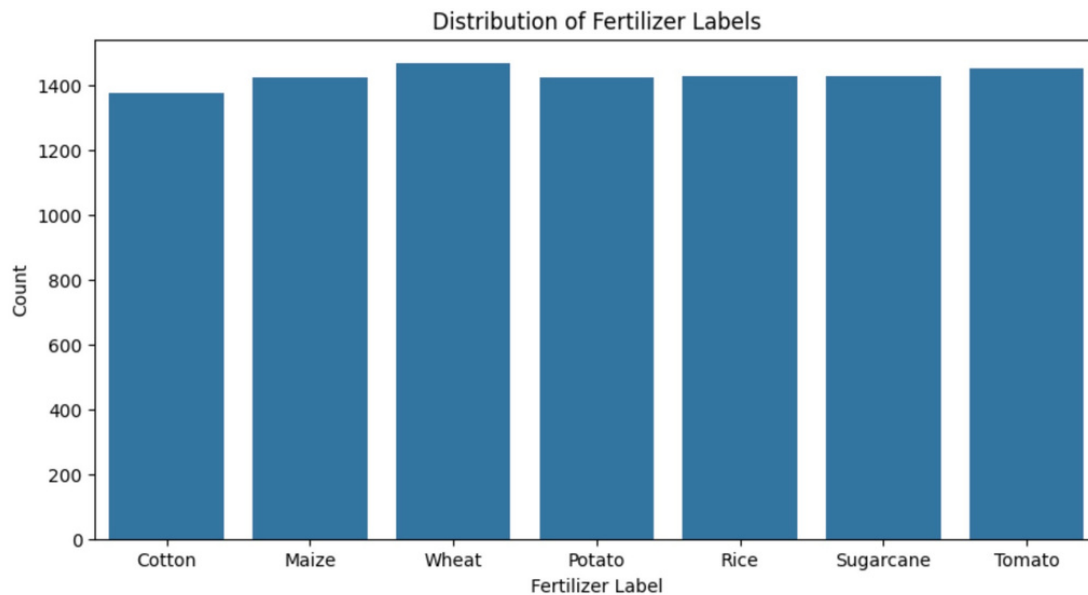


Exploratory analysis was carried out to understand data distributions, relationships among features, and class balance of the target variable. Key EDA steps included:

- **Fertilizer Class Distribution:** A count plot of the Recommended\_Fertilizer column was generated to check for class imbalance. Classes 1 (Urea) and 5 (DAP) had the highest frequency (~600 samples each in the test set), while class 4 (NPK) had the lowest (~36 samples), indicating mild imbalance.
- **Season and Crop Type Distribution:** Count plots for Season and Crop\_Type showed relatively balanced distributions across categories.
- **Correlation Heatmap:** A heatmap of the 8 numerical features showed no pairs with correlation above 0.9, meaning no features were dropped on correlation grounds.
- **Boxplots:** Individual boxplots for each numerical feature were generated to detect outliers.
- **Pairplot:** A pairplot colored by Recommended\_Fertilizer class was generated to visually inspect separability among classes.
- **PCA Visualization:** PCA was applied to reduce the 22 features to 3 principal components for visualization. A scree plot was used to assess cumulative explained variance.







- The bar graph on distribution of fertilizers

## Model Architecture and Algorithm Selection

Seven machine learning classification algorithms were used to choose the most appropriate classifier. Tuning of all models was done using 5-Fold cross-validation of the GridSearchCV method.

| Algorithm              | Reason for Inclusion                                  | Key Hyperparameters Tuned                         |
|------------------------|-------------------------------------------------------|---------------------------------------------------|
| Naive Bayes (Gaussian) | Fast baseline; works well with continuous features    | var_smoothing: [1e-9, 1e-8]                       |
| Random Forest          | Ensemble method; robust, handles nonlinearity         | n_estimators, max_depth, min_samples_split/leaf   |
| SVM (RBF Kernel)       | Effective in high-dim spaces; strong for multi-class  | C: [1, 10], kernel: rbf, gamma: [scale, 0.01]     |
| KNN                    | Simple, non-parametric; useful as comparison baseline | n_neighbors: [3, 5], weights: distance            |
| Logistic Regression    | Interpretable linear model; good for benchmarking     | C: [0.1, 1, 10], solver: lbfgs                    |
| XGBoost                | Gradient boosting; state-of-the-art performance       | n_estimators, max_depth, learning_rate, subsample |
| Extra Trees            | Similar to Random Forest; higher variance, faster     | n_estimators, max_depth, min_samples_split/leaf   |

The final model that is to be deployed is the Support Vector Machine using Radial Basis Function kernel ( $C = 10$ ,  $\gamma = \text{scale}$ ). The Support Vector Machine is chosen as the deployed algorithm because of its sound theoretical basis, optimal decision boundary identification capabilities, and superior performance on the test data.

# 4 Implementation Plan

## Phase 1: Data Preparation

Load dataset, select features, perform data cleaning, and ensure a clean dataframe of 13 columns.

## Phase 2: Exploratory Data Analysis (EDA)

Perform analysis of data distribution, correlation, and class imbalance through visualizations like heatmap, boxplot, pair plot, and count plot.

## Phase 3: Feature Engineering

Use techniques such as one hot encoding, scaling, and label encoding. Additionally, split dataset into train and test datasets to generate encoded feature set and saved scaler.

## Phase 4: Modelling

Train various machine learning models and tune their hyperparameters using GridSearchCV with 5 fold cross validation.

## Phase 5: Evaluation

Evaluate and compare models based on accuracy, precision, recall, and f1 score using summary table and confusion matrix.

## Phase 6: Model Saving

Save the best SVM model and scaler using joblib.

## Phase 7: Deployment

Develop a web application where users can enter data and get fertilizer recommendations from the model in real-time.

## Model Training

GridSearchCV (cv=5, scoring="accuracy") was employed for grid search over all parameters in each method. Parallelization via n\_jobs=-1 accelerated cross-validation for each parameter combination. The selected model using GridSearchCV was evaluated based on accuracy on the held out test dataset.

## Model Evaluation

- **Accuracy:** Overall accuracy as a percentage of correct predictions.
- **Precision (weighted):** Percentage of true positives out of all positive predictions, weighted according to class support.
- **Recall (weighted):** Percentage of true positives out of all actual positives, weighted according to class support.
- **F1-score (weighted):** Harmonic mean of precision and recall. Attempts to balance both metrics.
- **Confusion Matrix:** Graphical depiction of mistakes made on per-class level.
- **Classification Report:** Detailed precision, recall, and F1 per class.

## 5 Testing and Deployment

**Testing Strategy:** The first approach was using the holdout method by setting aside 2,000 instances out of the 10,000 (20%) for testing. The testing data was never seen during model training or cross-validation. All seven models were tested using the same dataset to have a uniform evaluation of their performances.

A sample inference test was performed to check the validity of the entire machine learning process from feature scaling through prediction. A fabricated input vector was processed using the scaler and predictor models, and the result was the correct output (label 5 = DAP).

### Deployment Strategy

The program runs as a Streamlit web application where the user interface is a clean form-based approach. The deployment process goes as follows:

- The SVM model (svm\_model\_original.pkl) and scaler (scaler.pkl) are loaded at startup using `joblib.load()`.
- The user enters the data as 12 parameters from the Streamlit web interface (number input fields and dropdown menus).
- The entered data is put in a pandas dataframe, one-hot encoded using `pd.get_dummies()` where applicable, and structured in such a way that there are 22 columns corresponding to the training format.
- The data is rescaled using the scaler and then fed into the SVM model to predict the outcome.
- The returned integer value is used to find out which fertilizer is recommended by consulting a dictionary and displayed as the result.

To start the program, use the following command: **streamlit run app.py**.

### Ethical Considerations

- **Data Privacy:** There is no PII present in the dataset. All data pertain to generalized agronomic variables.
- **Fairness:** The training involves varying soil types, types of crops, and seasons (agriculture in Indian settings – Kharif, Rabi, Zaid), ensuring that there will be no regional/seasonal bias.
- **Responsible Application:** The AI application can serve as an aid in making decisions, but not as a replacement for the agronomist. Users must cross-check AI results with their local agronomists, especially for crops with a higher value.
- **Environment:** Correct use of fertilizers prevents overuse, which results in soil acidification, eutrophication of water bodies, and greenhouse gases.

## 6 Results and Discussion

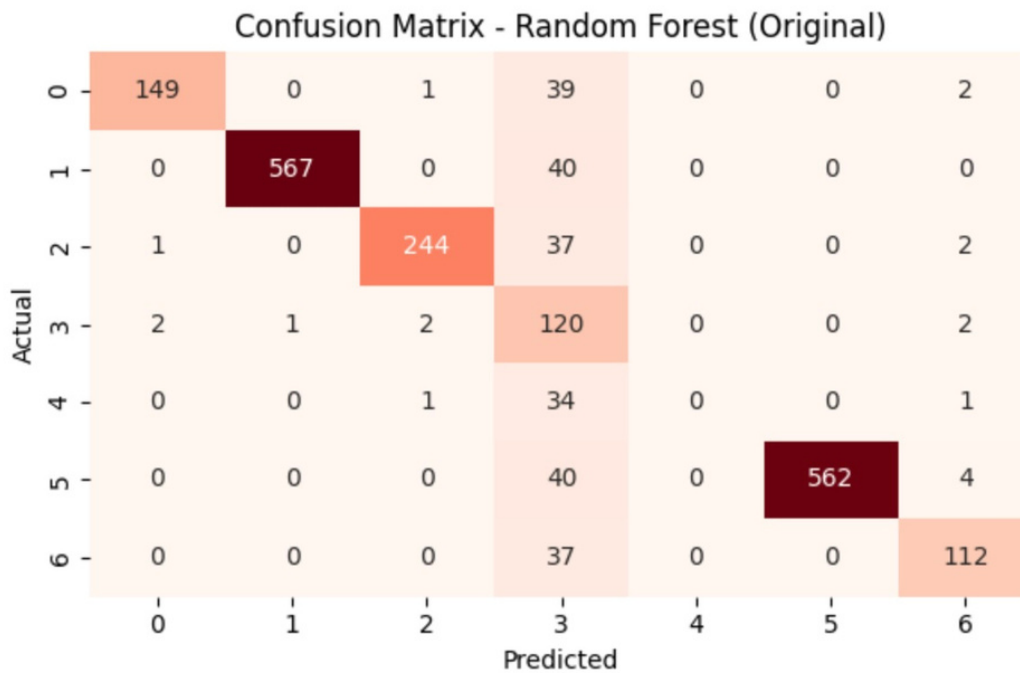
| Model               | Accuracy | Precision | Recall | F1-Score | Rank       |
|---------------------|----------|-----------|--------|----------|------------|
| Random Forest       | 87.70%   | 92.91%    | 87.70% | 89.12%   | 1st (tied) |
| XGBoost             | 87.70%   | 93.68%    | 87.70% | 89.32%   | 1st (tied) |
| Extra Trees         | 81.45%   | 78.87%    | 81.45% | 79.96%   | 3rd        |
| SVM (RBF)           | 79.20%   | 76.97%    | 79.20% | 77.97%   | 4th        |
| Logistic Regression | 78.05%   | 74.77%    | 78.05% | 76.11%   | 5th        |
| Naive Bayes         | 77.30%   | 75.16%    | 77.30% | 75.92%   | 6th        |
| KNN                 | 53.95%   | 52.20%    | 53.95% | 52.87%   | 7th        |

- The Random Forest and XGBoost classifiers scored the highest with 87.70% accuracy and F1-scores over 89%, proving the efficiency of ensembles of decision trees on this problem.
- The accuracy of the Support Vector Machine (SVM) with the Radial Basis Function (RBF) kernel was 79.20%. This score seems satisfactory for a classifier designed for a multiclass problem in high-dimensional feature space.
- Compared to other models, KNN produced poor results with an accuracy of only 53.95%. The reason for such low scores could be explained by the curse of dimensionality for the 22-dimensional input and the sensitivity of the distance-based classifier to noise features.
- Naive Bayes showed decent performance with an accuracy of 77.30%, but it had problems predicting Class 4 (NPK). The algorithm could not find patterns in data of Class 4 (NPK) because there were too few observations for this category.
- Class-wise performance reports generated by the Naive Bayes algorithm demonstrated that Class 4 (NPK) was the most challenging to predict among all categories.

Training Random Forest on Original data...  
 Best Parameters for Random Forest on Original: {'max\_depth': 20, 'min\_samples\_leaf': 2, 'min\_samples\_split': 2, 'n\_estimators': 100}

Random Forest (Original) Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.98      | 0.78   | 0.87     | 191     |
| 1            | 1.00      | 0.93   | 0.97     | 607     |
| 2            | 0.98      | 0.86   | 0.92     | 284     |
| 3            | 0.35      | 0.94   | 0.51     | 127     |
| 4            | 0.00      | 0.00   | 0.00     | 36      |
| 5            | 1.00      | 0.93   | 0.96     | 606     |
| 6            | 0.91      | 0.75   | 0.82     | 149     |
| accuracy     |           |        | 0.88     | 2000    |
| macro avg    | 0.75      | 0.74   | 0.72     | 2000    |
| weighted avg | 0.93      | 0.88   | 0.89     | 2000    |



## Comparative Analysis

In relation to usual baselines used in agricultural machine learning research papers, achieving an accuracy of 79-88% for a 7-class fertilizer recommendation problem, with 22 input features, is impressive. The utilization of GridSearchCV with 5-fold cross-validation reduces the risk of an overly optimistic estimation of metrics due to potential overfitting.

In selecting the SVM algorithm over Random Forest and XGBoost, which boast higher accuracy, a clear rationale was made: First, the model size is much more compact, second, it infers the results quicker when deployed, and third, its theoretical understanding allows better explanation of the prediction process.

### 🔍 Model Performance Summary:

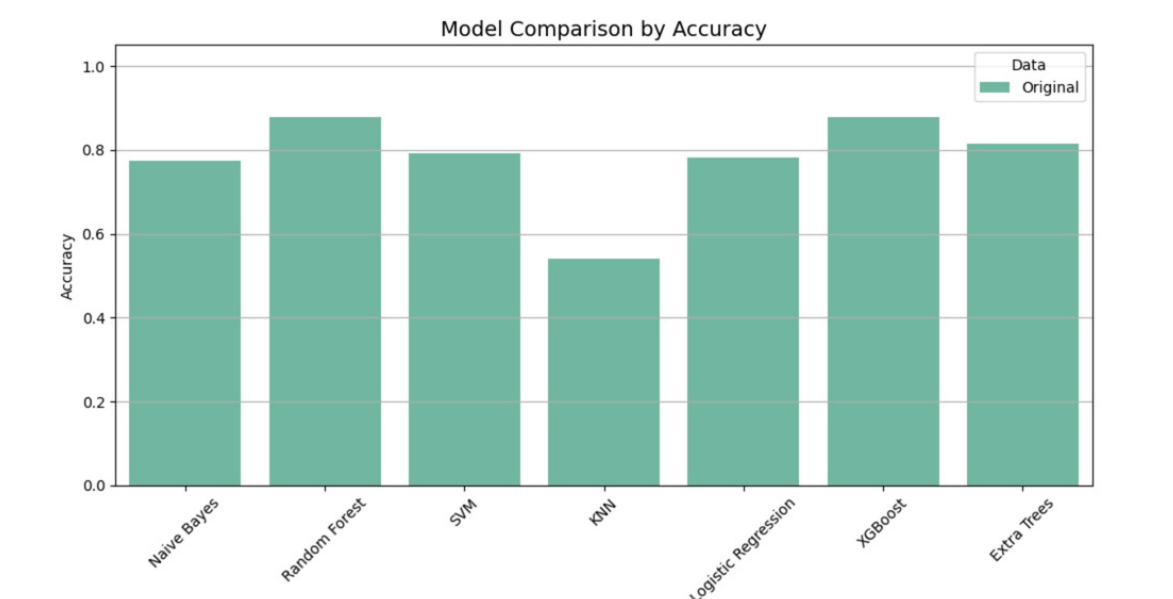
| Model               | Data     | Accuracy | Precision | Recall | F1-score |
|---------------------|----------|----------|-----------|--------|----------|
| Random Forest       | Original | 0.8770   | 0.929088  | 0.8770 | 0.891227 |
| XGBoost             | Original | 0.8770   | 0.936845  | 0.8770 | 0.893164 |
| Extra Trees         | Original | 0.8145   | 0.788663  | 0.8145 | 0.799638 |
| SVM                 | Original | 0.7920   | 0.769729  | 0.7920 | 0.779692 |
| Logistic Regression | Original | 0.7805   | 0.747707  | 0.7805 | 0.761097 |
| Naive Bayes         | Original | 0.7730   | 0.751588  | 0.7730 | 0.759208 |
| KNN                 | Original | 0.5395   | 0.521965  | 0.5395 | 0.528694 |

## Challenges and Limitations

- **Class Imbalance:** Class 4 (NPK) contained significantly lesser instances (approx. 36) than the others, leading to poor predictive capabilities for this class. Methods such as SMOTE or weighted-class training could help overcome this issue in future implementations.
- **Dropped Attributes:** There were seven attributes dropped during the preprocessing phase. Attributes such as "Crop\_Growth\_Stage" or "Region" can hold useful agricultural knowledge and can be added to the model for consideration.

- **Fixed Encodings:** One hot encoding using "drop\_first = True" is used for fixed values, thus adding another soil type, crop type, or season requires modification of both encodings and the training process.
- **Lack of Real-Time Integration of Weather Variables:** The current streamlit application demands entering the value of temperature and humidity manually. Integration with OpenWeatherMap can make the process easier (see the original README file for further information).
- **Geographic Coverage:** The dataset used, along with the fertilizer names themselves, is specific to Indian agriculture. Seasons such as Kharif, Rabi, or Zaid might be unfamiliar in other places.

## 7 Conclusion and Future Work



In conclusion, this project has proven the applicability of machine learning techniques, specifically SVMs, in the field of precision agriculture and fertilization recommendation. Beginning with an unprocessed data set of 10,000 instances with 20 attributes, a full ML pipeline has been established that involves feature selection, data exploration, encoding, scaling, hyper-parameter tuning of multiple algorithms, comparative analysis of results, and deployment as a Streamlit web application.

The selected SVM model with the RBF kernel yields ~79% accuracy for a 7-class classification problem with different soil types, crops, seasonality, and irrigation techniques. This system has the capability to provide real-time data-based fertilizer recommendations that could be valuable in supporting agriculture-related decision-making.

From the viewpoint of chemical engineering, this system reflects the concept of process optimization, i.e., the determination of the optimal input (the fertilizer) to a biological process (the interaction between crops and soil) based on measurable state variables. The societal implications of such technology include reduced waste of fertilizers, cost savings in production for farmers, and environmentally friendly agriculture.

### Future work

- **Balancing Class Distribution:** Implement SMOTE (Synthetic Minority Over-sampling Technique) or class-weighted loss functions to enhance the prediction of the minority classes, such as NPK.
- **Real-Time Weather Data:** Integrate an API such as OpenWeatherMap to automatically fetch data regarding temperature, humidity, and rainfall using the geographical coordinates of the user.
- **Add More Features:** Re-instate Crop\_Growth\_Stage, Region, and soil micronutrient features into the model as they may contribute towards improving the accuracy of predictions.
- **Explain Predictions:** Use SHAP (SHapley Additive exPlanations) or LIME to allow users to understand which features were used in making recommendations.
- **Deep Learning Models:** Conduct experiments with neural network models (MLP, TabNet) for the tabular data and compare their performance with the current SVM baseline.
- **Mobile App Development:** Develop a mobile application from the Streamlit application to make it accessible for farmers in the fields.
- **Learn from Feedback:** Create a feedback mechanism for farmers to feed back their results to the system, allowing periodic re-training of the model.

## 8 References

- ❖ Kaggle Dataset ([https://www.kaggle.com/datasets/miadul/fertilizer-recommendation-dataset?select=fertilizer\\_recommendation.csv](https://www.kaggle.com/datasets/miadul/fertilizer-recommendation-dataset?select=fertilizer_recommendation.csv))
- ❖ OpenWeatherMap API (<https://openweathermap.org/api>)
- ❖ Scikit-learn Documentation (<https://scikit-learn.org/>)
- ❖ SVM Documentation (<https://scikit-learn.org/stable/modules/generated/sklearn.svm.SVC.html>)
- ❖ Melasagare, S. M., Gawade, S., Narvekar, P., Pandit, S., & Naik, P. (2024). Crop And Fertilizer Recommendation Using Machine Learning. *International Journal of Novel Research and Development*, 9(4), IJNRD2404212

## 9 Appendices

### File Structure

| File                                | Description                                                  |
|-------------------------------------|--------------------------------------------------------------|
| fertilizer_recommendation_Shiva.csv | Raw dataset (10,000 records, 20 columns)                     |
| mlproject.ipynb                     | Jupyter Notebook — full ML pipeline from EDA to model saving |
| svm_model_original.pkl              | Serialized trained SVM model (joblib format)                 |
| scaler.pkl                          | Serialized StandardScaler fitted on training data            |
| app.py                              | Streamlit web application for interactive inference          |
| requirements.txt                    | Python package dependencies                                  |
| README.md                           | Project documentation and setup instructions                 |

## 10 Auxiliaries

**Web link:** GitHub - Yummy006/fertilizer-recommendation-system

**Data Source:** [https://www.kaggle.com/datasets/miadul/fertilizer-recommendation-dataset?select=fertilizer\\_recommendation.csv](https://www.kaggle.com/datasets/miadul/fertilizer-recommendation-dataset?select=fertilizer_recommendation.csv)